

Principles of Reliable Data Transfer

1 Introduction

Reliable data transfer (RDT) is a fundamental problem in networking. It appears at multiple layers, including:

- Link layer
- Transport layer
- Application layer

Service Model of Reliable Data Transfer

A reliable channel provides the following guarantees to the upper-layer entities:

- All data is delivered **without corruption**.
- No data is **lost**.
- All data is delivered **in order**.

This is the service model implemented by protocols like TCP.

2 Challenges

Implementing RDT is challenging because the underlying channel may be unreliable:

- Bits may be corrupted during transmission.
- Packets may be lost.
- Even though we assume packets are delivered in order, additional mechanisms are required to ensure reliability.

3 RDT Protocol Interfaces

We incrementally develop an RDT protocol, introducing the sender and receiver sides:

- `rdt_send()`: Called by the application layer to send data from the sender.
- `rdt_rcv()`: Called when a packet arrives at the receiver side.
- `deliver_data()`: Used by the receiver to pass data to the upper layer.
- `udt_send()`: Used by the RDT protocol to send packets over the unreliable channel.

Key Terminology

- **Packet:** Generic term for data unit transmitted over the channel.
- **Segment:** Transport-layer specific term (used in TCP).
- **Unidirectional transfer:** Data flows from sender to receiver only.
- **Full-duplex:** Bidirectional transfer, more complex but conceptually similar.

4 Unidirectional Data Transfer Model

Even in unidirectional transfer:

- The receiver may send **control packets** back to the sender.
- Both sender and receiver use `udt_send()` to communicate over the unreliable channel.
- This ensures mechanisms such as acknowledgments and retransmissions can operate correctly.

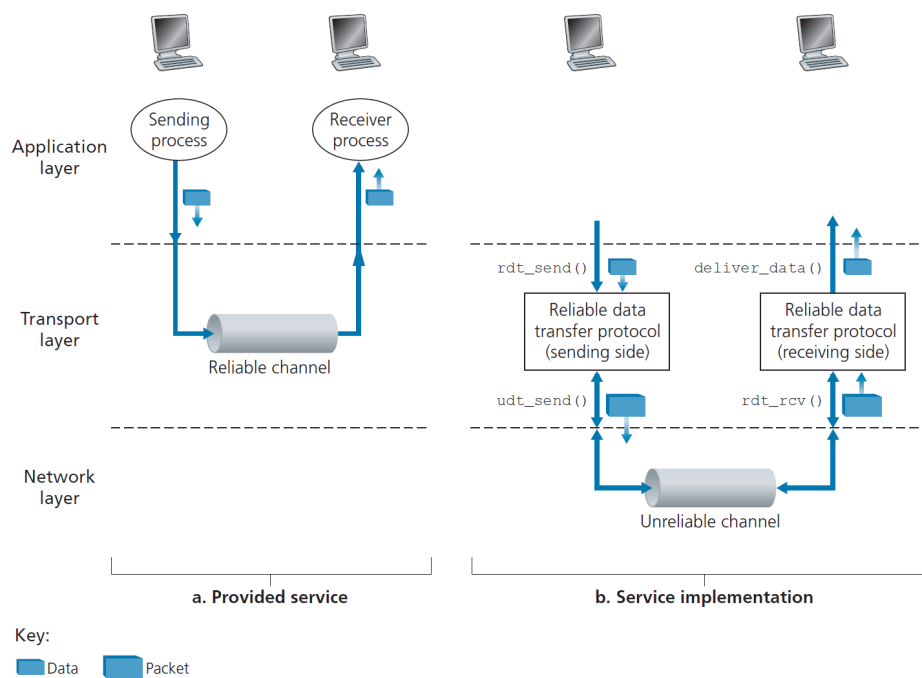


Figure 1: Reliable Data Transfer: Service model and implementation.

Reliable data transfer protocol (rdt): interfaces

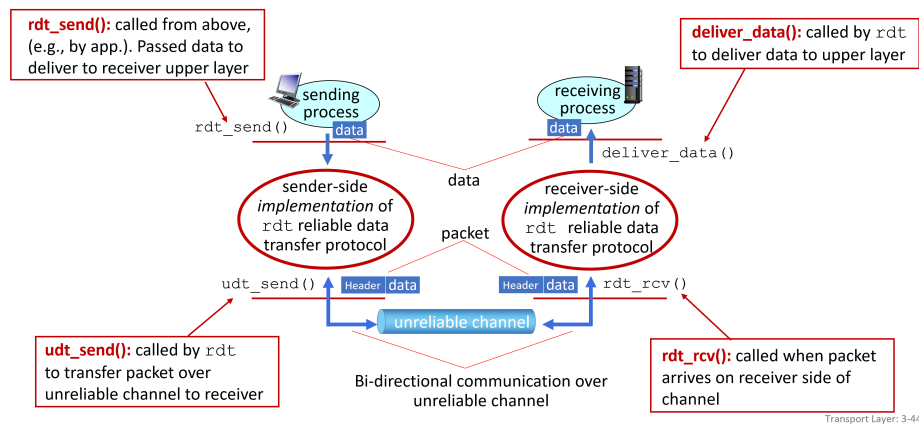


Figure 2: Reliable Data Transfer protocol (rdt): interfaces

5 Summary

Reliable data transfer protocols implement a reliable channel on top of an unreliable network. Key principles include:

- Error detection and correction
- Packet acknowledgment and retransmission
- Maintaining sender/receiver state
- Handling control packets for reliability

This framework forms the basis for protocols like TCP and is essential for robust network communication.